
TML26 - MEMBERSHIP INFERENCE ATTACK AS1

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1 INTRODUCTION

The goal of Membership Inference Attacks (MIA) is to determine whether a specific data sample was used in the training set of a target machine learning model. This can create a significant privacy risk because it may reveal sensitive information about individuals whose data was included in the training set.

In this task, participants are provided a trained ResNet-18 image classification model and must predict membership for an evaluation dataset. Submissions must assign a continuous score bounded between 0 and 1 for every sample, where a higher score indicates a higher likelihood of membership. Because both members and non-members are drawn from the same underlying distribution with no explicit membership indicators, the attack must identify small variations in the model's behavior on seen versus unseen data.

2 METHODOLOGY

2.1 ATTACK PRINCIPLE: THE META-CLASSIFIER APPROACH

We presented the Membership Inference Attack as a supervised binary classification problem rather than just relying on a single heuristic metric like the target model confidence or cross-entropy Loss. Although highly regularized models may not display a massive gap in overall loss, their internal representations (logits) still contain multidimensional signatures of memorization. Here, we used a "meta-classifier" approach. We can train a secondary machine learning model to learn the subtle decision boundary that separates members from non-members by extracting the raw output logits generated by the target model for each image.

2.2 FEATURE EXTRACTION AND TARGET MODEL

We used the provided pretrained ResNet-18 model, which is adapted for 32×32 images and 9 output classes. We evaluate the target model on the public dataset (`pub.pt`), which contains ground-truth membership labels to extract our training features. For each image, we apply the standard evaluation transformations (resizing to 32×32 and normalizing using the provided mean and standard deviation). We then perform a forward pass through the target model in evaluation mode to extract the 9-dimensional logit vector. This vector serves as the feature input (X) for our attack model.

2.3 ATTACK MODEL TRAINING (LOGISTIC REGRESSION)

With the extracted logits (X) and the ground-truth membership labels (y) from the public dataset, we trained a Logistic Regression classifier using the `scikit-learn` library. The model was configured with a maximum of 2000 iterations (`max_iter=2000`) to ensure proper convergence on the feature space. During inference on the hidden private dataset (`priv.pt`), the Logistic Regression model evaluates the 9-class logits and outputs a probability score using the `predict_proba` function. Because this function inherently bounds outputs to the $[0, 1]$ range, the predicted probability of Class 1 (Member) directly satisfies the assignment's scoring requirements.

2.4 DATA HANDLING

Some samples contained missing values or edge cases during the initial pipeline implementation. To ensure robustness and prevent runtime failures, we applied the following steps: (1) Missing images

were replaced with zero tensors to maintain batch integrity. (2) Missing labels and IDs were assigned default values to ensure no missing IDs in the final CSV. This guaranteed that every sample appeared exactly once in the final submission.

3 RESULTS

Our submissions are evaluated using the True Positive Rate at a 5% False Positive Rate (TPR@5%FPR)[cite: 3]. Our Logistic Regression meta-classifier achieved a leaderboard score of approximately:

$$\text{TPR@5\%FPR} \approx 0.056$$

Although modest, crossing the 0.050 threshold shows that the Logistic Regression model captures a meaningful membership signal beyond random guessing or simple loss-based methods.

4 DISCUSSION

The results show that the target ResNet-18 model is exceptionally well-regularized, exhibiting a very small generalization gap. Consequently, a simple loss-based thresholding approach effectively performs at random chance, as the loss distributions for members and non-members heavily overlap.

By utilizing Logistic Regression, we allowed an algorithm to dynamically weight the importance of different logit activations. This confirms that membership leakage is not always a single scalar value (like overconfidence in the true class), but can be a complex relationship across all predicted classes. Even in highly regularized models, the raw output distributions contain hidden topological fingerprints of the training data.

5 LIMITATIONS AND FUTURE WORK

The main limitation of this approach is its dependence solely on the target model’s outputs (a black-box setting) without establishing a baseline for how the model *should* react to unseen data. The Logistic Regression model can overfit to the public dataset if not carefully validated.

Future improvements to substantially increase the TPR@5%FPR score would involve transitioning to Reference-Based Attacks:

- **Shadow Modeling:** Training an identical ResNet-18 model on a known split of the public data to generate a baseline “unseen” loss distribution.
- **Likelihood Ratio Attacks (RMIA):** Comparing the target model’s loss directly against the shadow model’s loss to expose memorization, as proposed by Carlini et al. and Zarifzadeh et al. as given in the assignment pdf.

6 CONCLUSION

This work demonstrates that applying a Logistic Regression meta-classifier to a model’s raw logits can somewhat extract membership information, even when the target model is heavily regularized and conceals obvious overfitting metrics. However we noticed this is not a fool-proof technique and we were not able to get the performance up any further.

GitHub **Repository:**
Trustworthy-Machine-Learning

<https://github.com/CruelMarco/>